

Fletcher Porter

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I'm a jack-of-all-trades engineer with particular skill in mechanical design and software development. I love to get my hands dirty in every piece of a system. I find joy in having a broad understanding of everything and intimate knowledge of a few things I get to own.

Education

Aalto University · Ph.D Mechanical Engineering · Expected 2029

- Studying improving software tooling for the development of mechatronic systems
- Supervised by Prof. Jari Vepsäläinen

Aalto University · MS Mechanical Engineering · December 2024

- Wrote thesis “[A Model for Selecting Software Tools for Mechatronic Systems](#)”

University of California, Santa Barbara (UCSB) · BS Mechanical Engineering · Sept. 2019

- Exchange study in 2017–2018 at Lunds Tekniska Högskola in Lund, Sweden

English · Native speaker **Finnish** · B1

Certified SolidWorks Associate (CSWA) · Jan. 2020

Experience

Aalto University Fluid Power Lab · Doctoral Researcher, Research Assistant · Oct. 2022 – Present

- Developing an industrial automation framework using the Rust programming language
- Interviewing R&D engineers on how they reason about the quality of the electronic tools
- Developed control systems for industrial hydraulics with Beckhoff TwinCAT
- Integrated CAN and EtherCAT buses into a single system
- Developed communication with external software with TCP sockets

Oblong Inc. · Software Developer · Dec. 2020 – March 2022

- Qualified new hardware for a complex, embedded, multimedia, hardware/software device
- Collaborated on a hybrid remote/in-person team in the midst of the COVID-19 pandemic
- Improved logging on products to quicken Oblong's response to customer issues
- Debugged audio issues involving Linux, PulseAudio, Cisco hardware, and Oblong's hardware
- Took on maintainance and development of Oblong's legacy products and software tools
- Developed polyglot software in Python, Go, Bash, C++, C, Ruby, JavaScript, and TypeScript

NASA Jet Propulsion Laboratory · Robotics Intern · Summers 2018, 2016, and 2014

- Developed a system to study robotic mobility for missions to icy moons
- Designed in CAD a mechanical system to offload gravity with minimal lateral forces
- Prepared engineering drawings of mechanical components to send to vendors to machine
- Designed an electrical system to distribute power to microprocessors that control the system
- Assembled the mechanical and electrical systems by hand
- Wrote automatic control programs and documentation in Python and Arduino C
- Developed a proposal for [a probe to bore ~20 km into Europa's icy crust](#)

- Designed and executed a test to demonstrate the feasibility of the proposed system
- Created drawings for parts to manufacture and to get ROM quotes
- Co-authored “A deep subsurface ice probe for Europa” in IEEE Aerospace Conference 2017
- Designed components of a robot manipulator in CAD and iterated on the design

Tetra Bio Distributed · Software Engineer, Board Secretary · April 2020 – Jan. 2021

- Developed a software display to show respiration descriptors of COVID-19 patients to doctors
- Designed a web server to send patient data to a browser front end using Python and Go
- Developed a Continuous Integration system to automatically test the software
- Wrote drivers for a pressure and a flow sensor for an embedded Linux system
- Made signal processing tools to present sensor data to doctors
- Prepared IRS Form 1023-EZ to successfully get the organization 501(c)(3) non-profit status

Hawkes Group, UCSB · Undergraduate Researcher · March 2019 – Sept. 2019

- Iteratively designed in CAD a fixture to hold tools on the end of a soft, vine-like robot
- Built these fixtures by making them on a 3D printer and assembling them
- Developed models for jump height of a carbon fiber-bow jumping robot
- Worked under the supervision of Prof. Elliot Hawkes

Capstone Course Design Project, UCSB · Student Engineer · Sept. 2018 – June 2019

- Designed in SolidWorks from requirements a stair-climbing dolly that carries heavy payloads
- Machined most custom components by hand on a mill and lathe from drawings I made

UCSB Dynamic Robotics Laboratory · Undergraduate Researcher · Oct. 2015 – Sep. 2016

- Worked in the UCSB robotics lab with Prof. Katie Byl
- Developed roller skates for RoboSimian to significantly increase locomotive speed

Publications

Porter, Fletcher. “A Model for Selecting Software Tools for Mechatronic Systems”. MA thesis. Aalto University, July 2024. URL:

<https://aaltodoc.aalto.fi/items/d2e174be-633b-4896-a8c2-fa90887524a5>.

Porter, Fletcher and Jari Vepsäläinen. “A Model for Selecting Software Tools for Fluid Power Systems”. In: *The 12th JFPS International Symposium on Fluid Power*. Hiroshima, Japan: The Japan Fluid Power System Society, 2024, pp. 219–229.

Porter, Fletcher, Gert Kauniste, Oskari Hynninen, Yelin Hou, Panu Kiviluoma, and Petri Kuosmanen. “Development of a Robotic Fleet for Agriculture”. In: *Proceedings of the 8th Baltic Mechatronics Symposium*. Finland: Aalto-yliopisto, 2023. URL:

<https://research.aalto.fi/en/publications/development-of-a-robotic-fleet-for-agriculture>.

Wilcox, Brian H., Jason A. Carlton, Justin M. Jenkins, and Fletcher A. Porter. “A deep subsurface ice probe for Europa”. In: *IEEE Aerospace Conference*. 2017, pp. 1–13. DOI: [10.1109/AERO.2017.7943863](https://doi.org/10.1109/AERO.2017.7943863).

A portfolio of my work can be found at <https://blog.fletcherporter.com>